

Alejandro Gomez-Paz

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LinkedIn | GitHub | Website

Education

University of Washington

Expected June 2028

Bachelor of Science in Applied Mathematics: Data Science

GPA: 3.8

Coursework: Numerical Optimization (graduate), Bayesian Statistics, Statistical Reasoning, Data-oriented Python

Self-Study: Elements of Statistical Learning, Deep Learning

Experience

Sensors, Energy, and Automation Laboratory | Machine Learning Intern

June 2026 – Present

Built custom RAG pipeline for an embedded-AI product

Writeup

- Engineered a retrieval pipeline using a compute-optimal ranking algorithm in Python, reducing hallucination 72%
- Built a vector database with transformer-based embeddings, semantic-aware chunking, and optimal HNSW indexing across 30+ scraped sources, surfacing the correct source 29% more often
- Ran a parameter sweep over chunk size, retrieval depth, and index parameters to optimize retrieval quality under embedded memory/compute constraints, cutting tokens per query 34%
- Designed a 52-query evaluation framework derived from LLM research for objective validation

Laboratory Emissions Reporting System

- Automated live data acquisition from a REST API into PostgreSQL, sustaining 99% uptime
- Built a first-principles physics emissions model in SQL, propagating sensor measurement error into uncertainty bounds on every reported estimate
- Built an interactive timeseries dashboard with automated report generation in Looker/Data Studio, replacing 3 hours/week of manual reporting

Highlighted Projects

Explainable Fight Prediction Website for UFC

Live site | Writeup

A counter-factual classification model with confidence quantification and natural language reasoning served on a public website via automatic ETL pipeline

- Scraped, cleaned, and feature-engineered 3M+ datapoints using Pandas, including a hyperparameter-tuned rating system increasing accuracy by 14%
- Trained a logistic regression model on 10,000+ fights using scikit-learn, quantifying uncertainty with confidence intervals through a bootstrap ensemble
- Generated natural-language reasoning per prediction decomposed from model parameters
- Served from a public website backed by a self-updating ETL pipeline and cloud database with 100+ monthly visits

Research Discovery Engine | Finalist (Top 3 of 80+), UW WINFO Hackathon

Jan 2026

Abstract-grounded paper discovery over live Google Scholar, PubMed, and arXiv with auto-formatted citation export · 3-person team

- Engineered a custom MCP server in Python registering Google Scholar, PubMed, and arXiv search as LLM-callable tools, grounding every summary in a live abstract rather than model memory
- Built two-stage retrieval: bi-encoder embeddings to shortlist candidates, then a cross-encoder scoring query and abstract jointly, replacing the source APIs' keyword ordering
- Deduplicated overlapping hits across three repositories via normalized title matching in pandas, merging them into one unified ranked list
- Streamed results from an async FastAPI backend with query-level caching, rendering top matches before full retrieval completes

Skills

Languages & Tools: Python, SQL, Git/GitHub, LaTeX

ML & AI: PyTorch, scikit-learn, pandas, LLMs, RAG, vector databases

Data & Infrastructure: PostgreSQL, cloud databases, ETL pipelines, APIs, Looker/Data Studio